9 The bow is a powerful two-arm string machine used for archery. Fig. 9.1 shows the three types of bows, namely the Longbow, the Recurved and the Compound bow.

Longbow Recurved bow Compound bow

Fig. 9.1

Each bow consists of a limb and a handle as shown in Fig.9.2. The distance between the string and the bow handle at rest is known as the bracing distance. When a bow is drawn by the fingers of the archer, the string is not stretched but the shape of the bow is changed and bent and the string is displaced by a drawing distance. The shaft of the arrow is rested on the handle and the tail of the arrow is rested on the middle of the string.

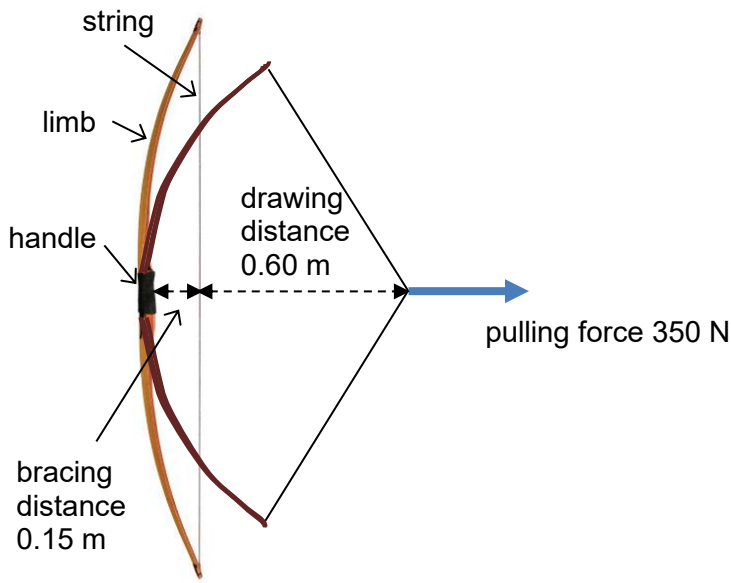


Fig. 9.2

In a Longbow, if we suppose that the bow is initially unstressed and the string is almost slack, then the archer starts to draw his arrow with a pulling force which is nearly zero and the pulling force increases with the drawing distance. The energy stored in the bow is equal to the work done in drawing back the string. In practice, a typical archer can draw an arrow back about 0.60 m and with a maximum force of about 350 N as shown in Fig. 9.3.

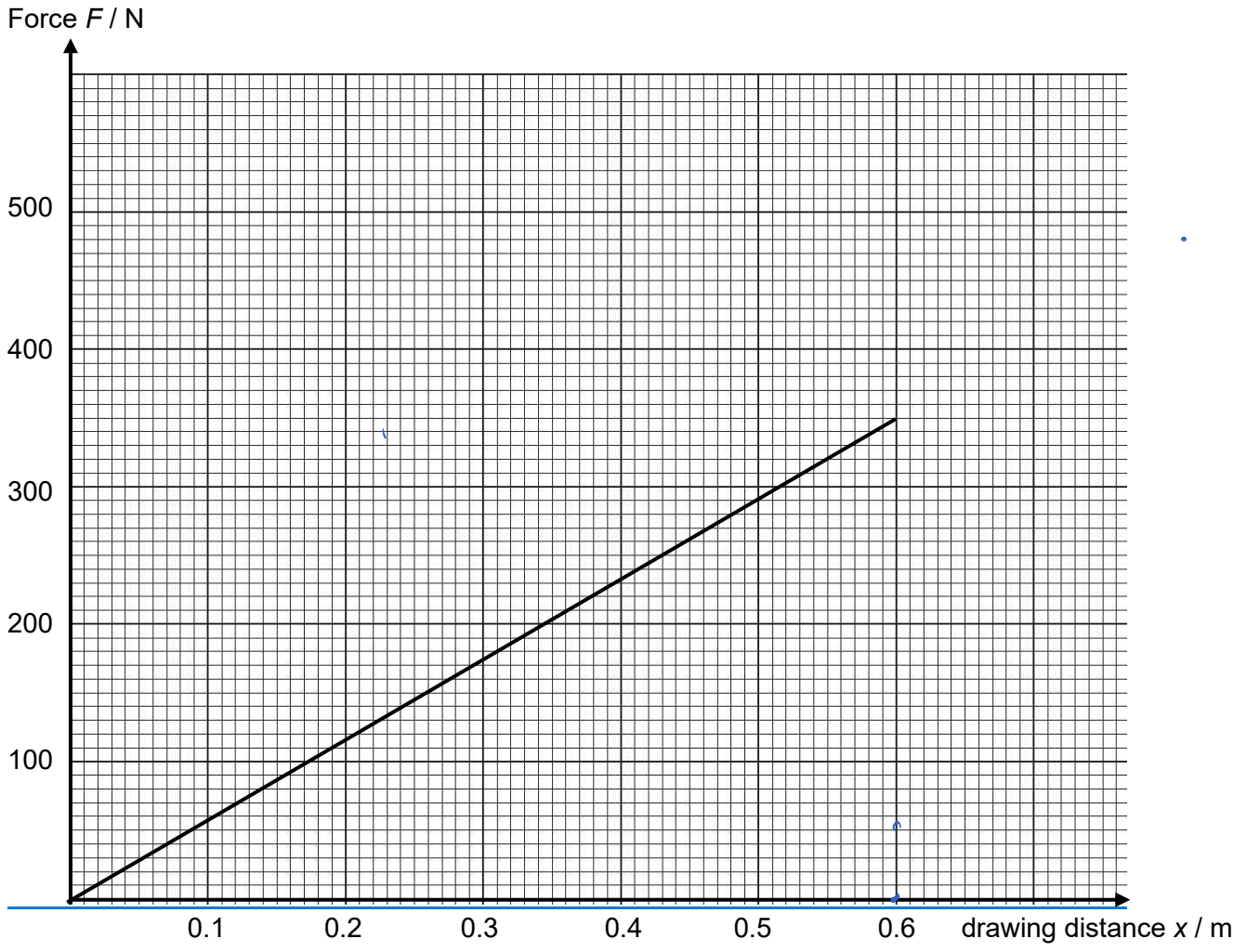


Fig. 9.3

The efficiency of the bow, η, can be defined as

$$\eta = \frac{\text{kinetic energy of the arrow}}{\text{elastic potential energy stored in the bow}}$$

When an arrow is shot, the force due to the tension in the string accelerates the arrow. A larger part of the energy stored in the bow is transferred to the arrow. Since this transferred energy is converted into the kinetic energy of the arrow $\frac{1}{2} m v^2$ (where v is the speed of the arrow as it leaves the bow), the increase in the length and therefore the mass of the arrow has two opposing effects: an increase in efficiency but a decrease in speed.

When a bow string is released, the string exerts a forward force on the arrow and causes it to accelerate forward. At the same time, there is a sudden compressive force along the length of the arrow, causing it to buckle. Hence the arrow will undergo **lateral vibrations** as it accelerates forward. Fig. 9.4 shows the top view of the arrow leaving bow (not to scale).

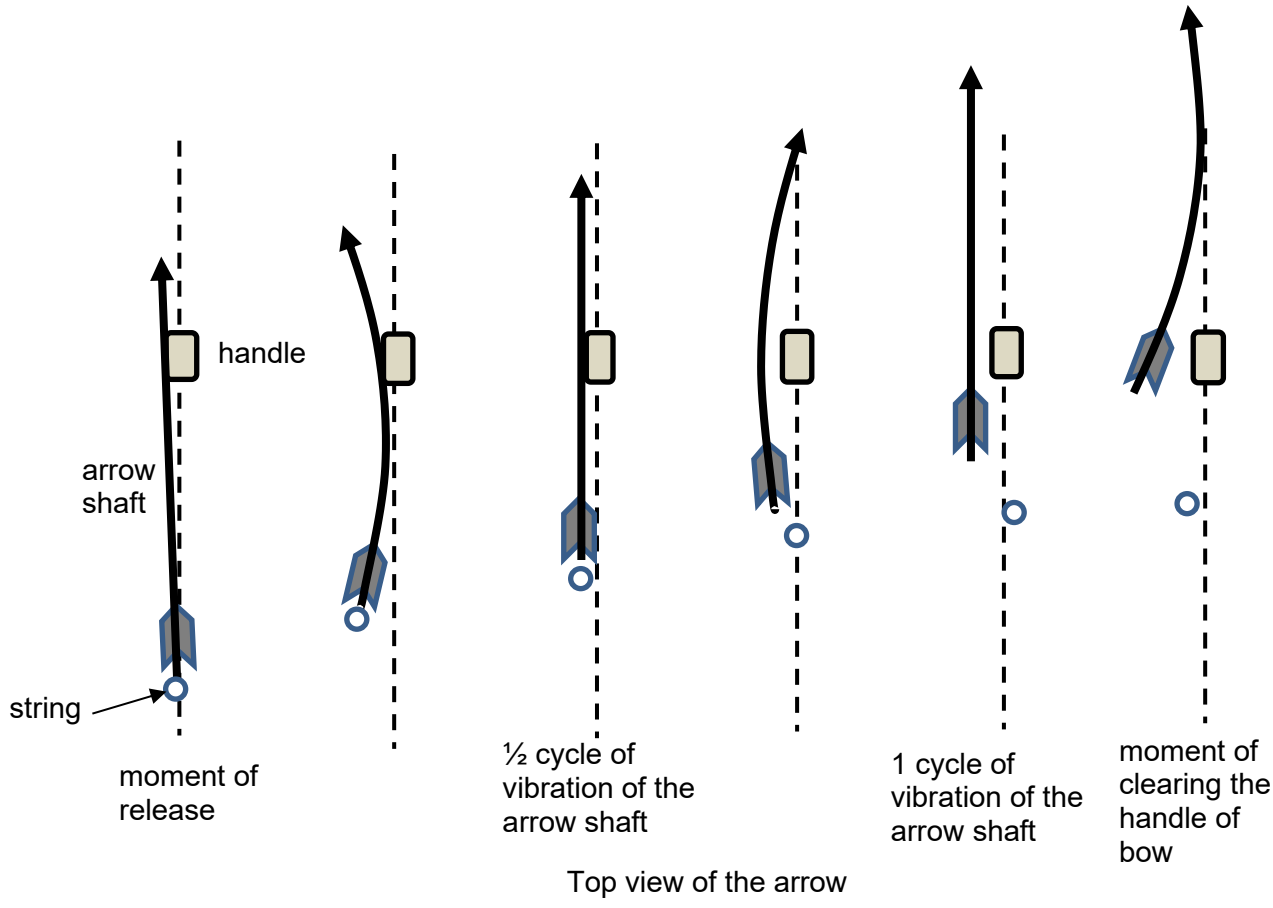


Fig. 9.4

Both the frequency and the amplitude of these vibrations must be matched to the bow if the arrow is to avoid hitting the side of the handle during its discharge. Ideally the arrow will make 1¼ vibrations from the moment of release until it finally clears the handle of the bow.

A Recurved bow is one in which the end of each limb curve away from the archer such that the limbs are braced with a bracing distance. The archer must start his pull with a non-zero force which is about 1/3 of the maximum force. When using a Recurved bow, the average force will be higher as compared to a Longbow without bracing.

In a Compound bow, which utilizes levers, the drawing force increases and decreases with the drawing distance as shown in Fig. 9.5.

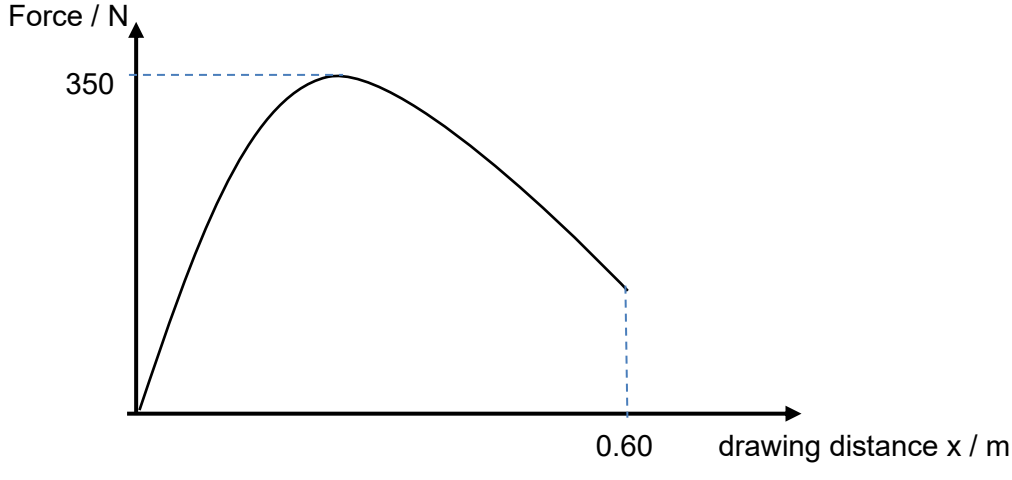


Fig. 9.5

(a) Explain what is meant by *lateral vibrations*.

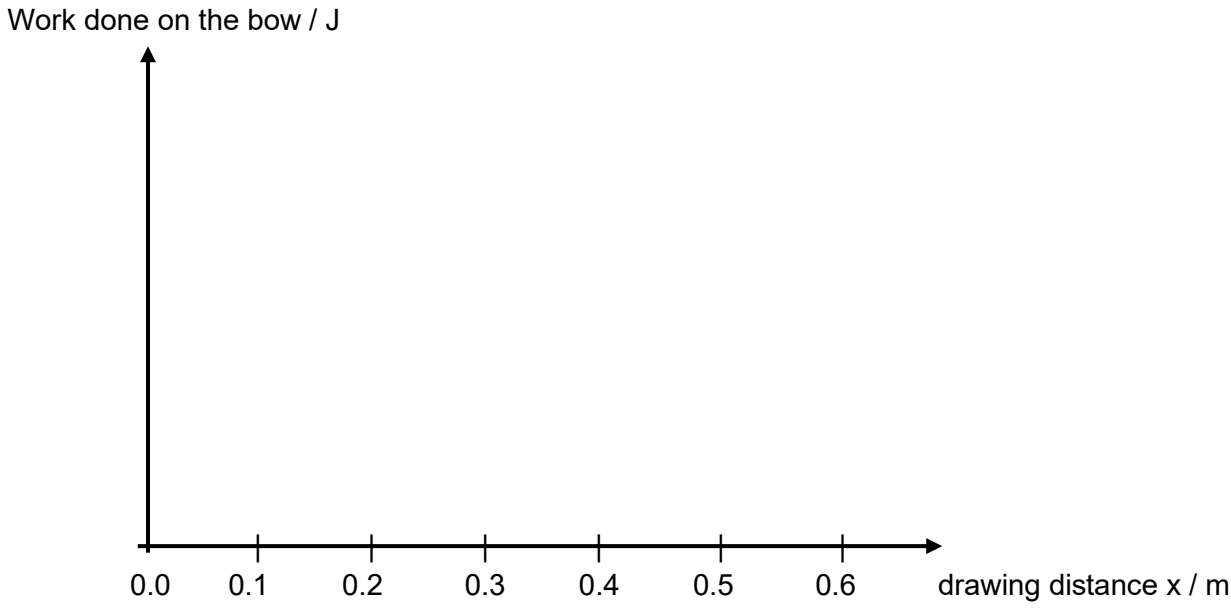
……..

…….. [1]

(b) Use the graph in Fig. 9.3 to calculate the energy stored in the Longbow when the maximum pulling force of 350 N is exerted on the string.

energy stored = ……J [2]

(c) On Fig. 9.6, sketch a graph to show how the work done on the Longbow changes with the drawing distance of the string. Label the work done axis clearly.



[2]

Fig. 9.6

(d) It is thought that the efficiency η of the bow obeys a relation of the form

$$\eta = \frac{m}{m + k}$$

where m is the mass of the arrow and
k is a constant depending on the mass of the bow.

A student performed an experiment to investigate how η, the efficiency of the bow varies with m, the mass of the arrow. He obtained data which allows him to plot the graph of Fig. 9.7.

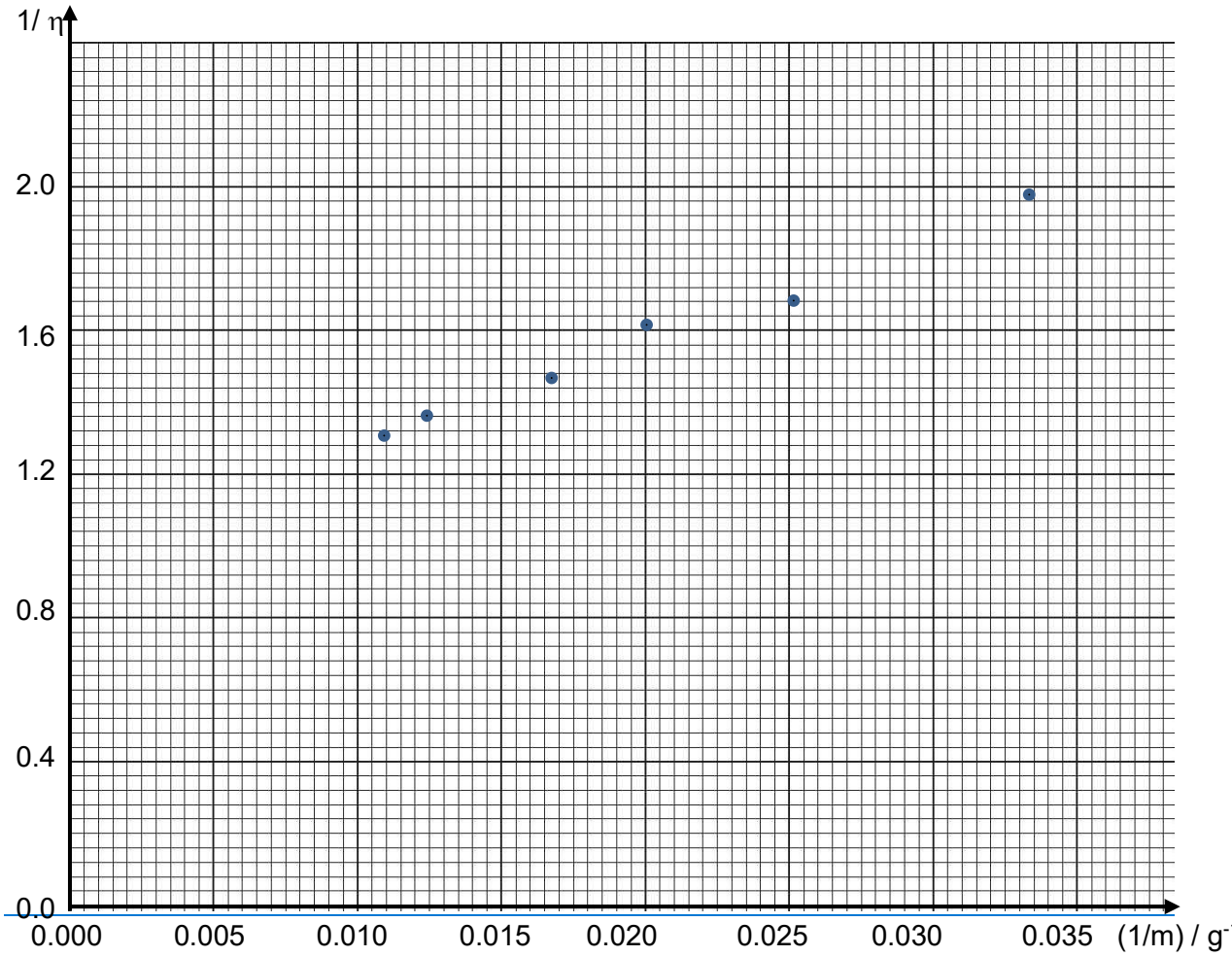


Fig. 9.7

(i) Draw the line of best fit for the points. [1]

(ii) Explain whether the relation in **(d)** is valid using the line drawn in Fig. 9.7.

……

……

……

……

…… [3]

(iii) Use the line drawn in Fig. 9.7 to determine the magnitude of the constant k in the expression in **(d)**.

k = …….. [2]

(iv) An arrow of mass 70 g is being shot from an initially unstressed Longbow drawn back as shown in Fig. 9.2. Use the graph in Fig. 9.7 and the definition of the efficiency to determine the speed of the arrow leaving the bow.

speed of the arrow = ……..m s^{-1} [3]

(e) Calculate the frequency of vibration for an arrow leaving the string at 50 m s^{-1}, from a bow of bracing distance of 0.15 m, and shot by an archer with a drawing distance 0.60 m, as shown in Fig. 9.2.

frequency of vibration = …….. s^{-1} [2]

(f) **(i)** On Fig. 9.3, sketch the force-drawing distance graph for a Recurved bow which is already braced by 150 N. The maximum drawing force of 350 N is exerted on the string at the maximum drawing distance of 0.60 m. [1]

(ii) State and explain, in terms of the energy stored, why the Recurved bow is better than a Longbow.

……

…… [1]

(g) Refer to the force-drawing distance graph of a Compound bow as shown in Fig. 9.5. Suggest an advantage of a Compound bow as compared to the Longbow.

. ……

…… [1]